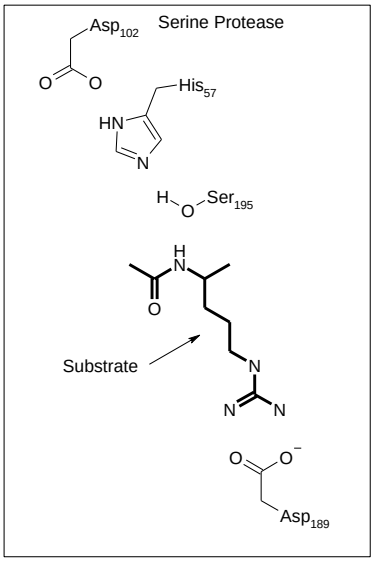
Biochemistry Cume

MSLLTEVETPTRNGWECVDGLEYWKSRTLLVALGIAIAMEMGVTELIRRVFIFSWLVTTLI
AVPLLWILDRLNERGHCGYHISIIGILHLILWILDRLNERGHCGYHISIIGILHLILWILD
RLNERGHCGYHISIIGILHLILWILDRLNERGHCGYHISIIGILHLILWILDRLNERGHCG
YHISIIGILHLILWILDRLNERGHCGYHISIIGILHLILWILDRLNERGHCGYHISIIGIL
HLILWILDRLNERGHCGYHISIIGILHLILWILDRLNERGHCGYHISIIGILHLILWILD

1. (30pts) The M2 protein is known for its role in facilitating the acidification of the viral interior, allowing the virus to escape from endosomes and enter the host cell's cytoplasm. This is a critical step in the viral replication cycle. Previous studies demonstrated that a stretch of 18 hydrophobic residues can adopt a transmembrane helix. Given the sequence, identify the stretch adopting transmembrane helix.
2. (30pts) You isolated an octapeptide. Subsequently, you obtained the following data: (1) Amino acid composition: 2K, G, M, F, H, S, E; (2)Treatment with chymotrypsin yielded a dipeptide and a hexapeptide; (3)Treatment with CNBr had no effect; (4) Treatment with trypsin yielded a tripeptide, a tetrapeptide, and free Met; (5) Treatment with dabsyl chloride yields dabsyl-Ser; (6) Treatment with Staphylococcal protease yielded a pentapeptide and tripeptide with no charge at pH 5. What is the sequence of the polypeptide?
3. A schematic diagram of trypsin active site is shown below. Describe the role of each



residue in hydrolysis of a substrate. What is the role of Ser195 (10pts)? What is the role of Asp102 (10pts)? What is the role of His57 (10pts)? What is the role of Asp189 (10pts)?